Dinushantha Meeriyagalla

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Research Profile

Human-Computer Interaction researcher and MSc Cognitive Systems candidate with experience designing, implementing, and evaluating interactive systems in virtual reality, automated driving, and immersive simulation. Combines experimental HCI methods with Unity/C#, Python, user-centered design, and professional software-development experience.

Research Interests

Immersive HCI and VR/XR; human-vehicle interaction and automotive HCI; human factors, cognitive load, and workload; interactive simulation; trust calibration and human-AI interaction; adaptive interfaces; physiological and behavioral measures in HCI.

Education

MSc Cognitive Systems Oct 2023 – Present
Ulm University, Germany

- Focus on Human-Computer Interaction, cognitive systems, interaction design, neurotechnology, and human-centered intelligent systems.
- Selected coursework: Foundations and Concepts of Cognitive Systems Modeling (1.0); Neurotechnology: Brain-Machine Interfacing (1.0); Cognition in Complex Situations (1.3); User-Centered Design for Interactive Systems I & II (1.3); Design Thinking for Interactive Systems (1.3).

BSc (Hons) Software Engineering Mar 2018 – Jul 2021
University of Plymouth, United Kingdom

- Second Class Honours, Upper Division; foundation in software engineering, web development, databases, distributed systems, and HCI.

Chinese Language Foundation Course Sep 2016 – Jul 2017
Nanjing Normal University, China

Master's Thesis

Comparison of Vehicle Motion Simulation in Virtual Reality Across the Motion-Fidelity-Continuum
Apr 2026 – Present
Ulm University | Human-Computer Interaction and Human Factors

- Designing an ongoing within-subject study comparing stationary VR, a YawVR motion platform, a room-scale VAMPIRE Chair condition, and a real-car reference condition.
- Developing and integrating Unity/C# experimental systems for vehicle-motion simulation, human-vehicle interaction, motion-fidelity comparison, and synchronized study procedures.
- Preparing measures of motion perception, presence, comfort, workload, and overall simulator experience, together with secondary interaction tasks and structured questionnaire and behavioral data collection.
- Current status: experimental design and system implementation are ongoing; participant results and final analyses are not yet reported.

Selected Research Experience

Trust Calibration and Adaptive Visualization in Automated Driving 2025 – Present

- Developed a Unity-based experimental simulation for studying trust, mental demand, and user perception in automated-driving scenarios.
- Integrated questionnaire feedback with an adaptive visualization workflow using Bayesian optimization concepts.
- Completed data collection for a 2×2 within-subject study with 24 participants; analysis remains ongoing.
- Online study: Visualization of Automated Vehicle Functionalities

Assessing Cognitive Load in Immersive Environments 2025

- Reviewed subjective workload measures and physiological and behavioral indicators used to assess cognitive load in immersive environments.
- Produced a research report connecting cognitive-load measurement, immersive HCI, user-study design, and human-centered evaluation.

Unity Prototype for Trust Calibration in Highly Automated Driving 2024 – 2025

- Built a Unity prototype and planned a 3 × 3 experimental structure combining three visualization approaches with three reliability levels.
- Implemented pedestrian-crossing, false-positive, false-negative, occlusion, trajectory-cue, and segmentation-highlighting scenarios.
- Prototype and study-design implementation were completed; a participant study was not conducted in this version.

Binaural Audio and Motion-Based Experimental Automation 2024 – 2025

- Contributed to automated audio playback, recording, motion tracking, experimental timing, structured data storage, and spectrogram generation for controlled cognitive and auditory experiments.

Research Methods and Technical Skills

Research methods: Experimental design; human-subject study execution; questionnaire-based evaluation; usability evaluation; cognitive-load and workload assessment; literature review; structured study procedures.

Experimental systems: Unity-based experiment development; VR/XR prototyping; secondary-task integration; hardware-software integration; experimental logging and data-collection workflows.

Programming and data: C#, Unity, Python, JavaScript, TypeScript, HTML, CSS, PHP, SQL; data preprocessing and cleaning; CSV/Excel workflows; structured data storage; basic signal-processing and data-visualization workflows.

HCI and design tools: User-centered design, UX research, interaction design, accessibility, wireframing, prototyping, Figma, Adobe XD, TYPO3, WordPress, Git, GitHub, and VS Code.

Academic and University Experience

Student Assistant May 2026 – Present
Ulm University, Executive Support Unit: Quality Development, Reporting and Revision

- Develop and maintain TYPO3 pages following institutional communication and content standards.
- Contribute to Python-based automation, data preparation, validation, and reporting workflows.

Student Assistant – Marketing Oct 2025 – Present
Ulm University, Central Student Advisory Service

- Design and maintain student-facing TYPO3 content, improving information structure, readability, user journeys, and bilingual presentation.

Student Assistant – Web Development May 2024 – Present
University Hospital Ulm, MOCA Lab

- Maintain research-related WordPress and TYPO3 websites and improve responsive layouts, performance, and alignment with research communication goals.

Student Assistant – Web Development Jan 2025 – Dec 2025
Ulm University, i3R Center

- Developed TYPO3 pages and frontend components for research and institutional communication, with attention to accessibility, usability, and performance.

Relevant Professional Experience

Front-End Engineer Nov 2021 – Nov 2023
WSO2

- Developed responsive and accessible interfaces, landing pages, websites, and marketing platforms using HTML, CSS, JavaScript, jQuery, and Bootstrap.
- Designed and prototyped web and email experiences in Figma and Adobe XD; contributed to information architecture, usability-oriented improvements, and A/B testing using Hotjar, GA4, and Pardot.
- Collaborated across design, engineering, and marketing teams in an agile environment.

Marketing Design Intern May 2021 – Sep 2021
WSO2

- Created reusable design assets and mobile-friendly email templates to improve visual consistency and user flow.

Selected Academic Writing

- Seminar paper: *Critical Analysis and Reflection of Concept Representation in Language Models.*
- Research project report: *Assessing Cognitive Load in Immersive Environments.*

Awards and Languages

Awards: Top 5% Global, WSO2 Performance Review (2022 H1); Top 20% Global, WSO2 Performance Review (2022 H2); National Rank 39 in Arts Stream, GCE Advanced Level Examination, Sri Lanka (2015).

Languages: English – Fluent; Sinhala – Native.